Reinaldo Aguiar Marcano
2302 Britton Ridge Drive, Katy TX 77494
https://xislist.org/ (ISBN 9781971481296)

# FORMAL NOTICE OF PROPRIETARY RIGHTS, INITIATION OF PROVISIONAL RIGHTS PERIOD, AND DEMAND FOR DUE DILIGENCE CONCERNING QUANTUM-CENTRIC SUPERCOMPUTING AND HYBRID CLOUD ARCHITECTURES

**This notice:**
https://storage.googleapis.com/rj-aguiar/IP_LIABILITY_QUANTUM_COMPUTING_COMPANIES.pdf

**TO: CHIEF EXECUTIVE OFFICERS AND CHIEF LEGAL OFFICERS**

akrishna@us.ibm.com (Arvind Krishna, Chairman and CEO, IBM Corporation)

michelle.browdy@ibm.com (Michelle H. Browdy, Senior Vice President, Legal and Regulatory Affairs, and General Counsel, IBM)

satya.nadella@microsoft.com (Satya Nadella, Chairman and CEO, Microsoft Corporation)

hossein.nowbar@microsoft.com (Hossein Nowbar, Chief Legal Officer, Microsoft Corporation)

**RE:** ACTUAL KNOWLEDGE OF PENDING PATENT CLAIMS (35 U.S.C. § 154(d)) REGARDING QUANTUM-CENTRIC SUPERCOMPUTING, DISAGGREGATED HYBRID ARCHITECTURES, AND THE AGUIAR MARCANO INTELLECTUAL PROPERTY PORTFOLIO

**DATE:** April 17, 2026

**DEAR:**

Mr. Krishna, Ms. Browdy, Mr. Nadella, and Mr. Nowbar,

This comprehensive technical and legal brief is served upon the executive leadership and chief legal officers of IBM Corporation and Microsoft Corporation to establish formal, legally binding actual notice.

As the architects of the world's most advanced classical and quantum computing infrastructures, your organizations are racing to achieve "utility-scale" quantum supremacy. We have closely monitored IBM's March 12, 2026 release of its "quantum-centric supercomputing" reference architecture, as well as Microsoft's recent unveiling of the Majorana 1 topological processor and its collaboration with the State of Maryland and the NSA to build the "Capital of Quantum" research center.

While your respective quantum processing units (QPUs)—whether superconducting transmons or topological Majorana zero modes—rely on differing physics, your overarching system architectures share a fatal legal vulnerability. The integration of these QPUs with classical CPUs and GPUs into unified, high-performance hybrid clouds fundamentally relies upon the proprietary claims of the



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Aguiar Marcano Intellectual Property Portfolio.

This portfolio comprises fifty-two (52) foundational patent applications formally executed and filed with the United States Patent and Trademark Office (USPTO).[1]

The primary objective of this document is to satisfy the strict requirements for "Actual Notice" pursuant to 35 U.S.C. § 154(d), thereby initiating the immediate accrual of provisional rights and the corresponding liability for reasonable royalties.[2]

# I. IBM Corporation: The "Quantum-Centric Supercomputing" Blueprint

**Target Operations:** IBM Quantum, Qiskit v2.1, and the QCSC Reference Architecture

On March 12, 2026, IBM published a comprehensive blueprint detailing the integration of QPUs into modern supercomputing environments, explicitly showcasing how quantum hardware operates alongside co-located CPU and GPU clusters connected via low-latency interconnects (e.g., NVQLink, RDMA). This exact blueprint is legally encumbered by our portfolio.

- **A Physically Disaggregated Computing System (USPTO 19/562,845) & System for a Data Center Architecture with Functionally Segregated Resource Pools (USPTO 19/562,855):** IBM's multi-level hardware infrastructure specifically relies on isolating the quantum system (which requires cryogenic cooling and specialized ASICs for error correction) from the "partner scale-up" and "scale-out" classical GPU/CPU clusters. Our patents explicitly claim this architecture: a computing system where primary components are physically disaggregated into separate enclosures, functionally segregated by specialized infrastructure zones, and interconnected by an ultra-high-bandwidth data fabric.[1]
- **Method for Dynamic Composition of Bare-Metal Computing Resources (USPTO 19/562,853):** IBM's system orchestration layer utilizes tools like the Quantum Resource Management Interface (QRMI) and Slurm plugins to expose quantum resources as schedulable entities. This directly infringes upon our method for dynamically composing a logical, bare-metal computer from physically disaggregated pools of hardware resources on demand.[1]

# II. Microsoft Corporation: Azure Quantum and Topological Scaling

**Target Operations:** Azure Quantum, Majorana 1 Processor, and Quantum-Classical Orchestration

Microsoft's roadmap hinges on scaling topological qubits and providing a hybrid developer environment through Azure Quantum and the Q# programming language.

- **Optical Computing Architecture and System (USPTO 19/560,038) & Massively Parallel**



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**Optical Data Bus Architecture (USPTO 19/562,818):** As Microsoft attempts to scale the Majorana 1 chip from a prototype to a system containing one million reliable qubits, the I/O bottleneck of moving classical control and error-correction data into and out of the dilution refrigerator becomes the primary physics barrier. Overcoming this requires abandoning serial electronics for massively parallel optical data fabrics. Any deployment of high-radix optical interconnects to bridge your topological cores with Azure's classical supercomputing backbone operates directly within our claimed hardware architecture.[1]

# III. The National Security Imperative and Federal Defense License

Both IBM and Microsoft are primary contractors for the United States Department of Defense, the Department of Energy, and the intelligence community. Microsoft's recent "Capital of Quantum" initiative specifically highlights its proximity to and collaboration with the NSA.

You must be advised that within the legal text of these patent applications, the inventor has executed an irrevocable, perpetual, worldwide, no-fee license granting the United States Federal Government the right to utilize these inventions for all national defense purposes.[1] Consequently, while the U.S. government is immune from infringement under 28 U.S.C. § 1498, your unauthorized commercialization of these architectures for private enterprise cloud offerings (e.g., commercial Azure or IBM Cloud clients) constitutes unshielded infringement.

# IV. Mandatory Due Diligence Protocols and Ultimatum

The statutory clock for the calculation of retroactive reasonable royalties commences immediately upon your receipt and actual knowledge of the published claims detailed herein. Ignorance of how your hybrid quantum-classical orchestration middleware and disaggregated hardware layouts synergize to infringe upon these overarching system-level claims is not a viable legal defense.

The executive leadership of each addressed organization is hereby directed to execute the following immediate remediation protocols:

1. **Initiation of FTO Audits:** You must immediately instruct your internal legal departments and outside IP counsel to cross-reference the fifty-two (52) applications within the Aguiar Marcano Portfolio against IBM's Quantum-Centric Supercomputing architecture and Microsoft's Azure Quantum hybrid workflows.

2. **Quarantine of Intersecting Architectures:** Any commercial deployment of disaggregated resource pools bridging QPUs and classical GPUs via high-bandwidth interconnects must be evaluated for immediate quarantine to mitigate ongoing commercial liability.



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We formally invite your respective legal departments to initiate a technical review of the cited USPTO applications. A formal response outlining your intended compliance audit, a presentation of non-infringement theories, or a request to discuss a commercial licensing framework is expected within thirty (30) business days of the transmission of this notice.

Govern yourselves accordingly.

Reinaldo Aguiar Marcano

Principal, https://xislist.org/

## Works cited

1. 02__ISBN-9781971481296__Portfolio-of-Patent-Applications_and_Declaration-D132.pdf
2. Houston, We've Got a [Provisional] Problem: An Analysis of 35 U.S.C. § 154(d), accessed April 9, 2026, https://scholarship.shu.edu/cgi/viewcontent.cgi?article=1126&context=student_scholarship



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